MECANO
THE MECHANICS OF CANON FORMATION

From Annotation to Interpretation: Exploring Pliny the Elder’s *Naturalis Historia* through Semantic Linking

Digital Humanities for Greek and Latin Linguistics and Philology

Sunoikisis DC Digital Scholarly Editions in Classics, Byzantine, and Medieval Studies: Session 7

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https://mecano-dn.eu

From digital annotation to text interpretation: where to start?

The case of the *Naturalis Historia*

Introduction

Why digitally annotating a Latin corpus?

- Semantic annotation allows to easily **extract information** about entities mentioned in the text
- Linguistic annotation enables the analysis of the **linguistic features** of a big corpus
- Together they offer the chance to analyze previously debated scholarly issues from a new angle with a **distant reading approach**

Introduction

The case of Pliny the Elder's *Naturalis Historia*

- **Many-voiced work** (Murphy 2004): many people are mentioned as sources, witnesses or protagonists of episodes
- Exploring the mentions of the individuals cited in the Latin encyclopedia in their **textual context**
- Understanding the mechanics of the selection performed by the author
- Focus on books 2-6 (geographical and astronomical topics)
- MECANO doctoral network: https://mecano-dn.eu/.

Data and Methodology

Implementing a digital pipeline

Step 1
Creating a high-quality digital corpus

When performing semantic annotation, it becomes crucial to work with a **high-quality digital text**

__Optical Character Recognition (OCR)__: automatic conversion of images containing text into machine-readable format

Mayhoff's critical edition (1892)

- OCR generated (Perseus Digital Library)
- Tokenized and lemmatized (Clérice 2021)

__Tokenization__: breaking down the text into smaller pieces (e.g. words)
__Lemmatization__: linking tokens to their own citation forms (lemmas)

Text quality assessment

Philological level

Formatting level

Step 1
Creating a high-quality digital corpus

Correction of the text:

- **Formatting** level: correction of the OCR errors
- **Philological** level: integration of different variant readings printed in Conte’s edition (Barchiesi et al., 1982)
- Text stored in a **relational database**: data stored in one or more tables, linked between each other
- Exploiting relational database **unique ids** to keep the text **compatible** with the original version
- Always possible to retrieve the original version and to distinguish between different types of modification

Step 2
Enhancing the text with automatic annotation

Part-of-Speech tagging: *assigning grammatical categories (e.g. verb, noun…) to tokens*

Linguistic annotation

New lemmatization and Part-of-Speech tagging

Automatic Annotation: how to find the best model?

- Trying with different models and evaluating the performance on a sample of 1000 tokens to choose the model with the best performance
- **Collatinus model:** trained on the LASLA corpus (classical Latin, manually annotated), and enriched with LatinBERT (Verkerk 2024)

Automatic annotation

Step 2
Enhancing the text with automatic annotation

Named Entity Recognition (NER)

*<u>NER:</u> **a**utomatic identification and classification of proper names in running text into predefined categories such as Persons, Places, Organizations.*

- Annotation of **persons**, defined as "any identifiable single individual" (Palladino et al. 2024)
- Definition of project-specific annotation guidelines
- Use of the **LatinBERT model fine-tuned** on this specific task (Beersmans et al., 2023)
- **Results correction** and **model evaluation**:
  - Bad precision due to the high number of populations and places mentioned only in this text: a lot of entities misclassified as persons
- Use of the linguistic annotation and the corrected OCR to speed and smooth the correction process

Semi-automatic annotation

Step 3
The linking: Wikidata and the Realencyclopädie

Wikidata

- Free, open and collaborative knowledge base
- Exploitation of structured data (e.g. M. Varro → citizenship → Ancient Rome)
- Enhancement of the coverage of the knowledge base with respect to the ancient world

***Realencyclopädie** der Classischen Altertumswissenschaft* (*RE*)

- Fundamental prosopographical reference work for the ancient world
- Entities extracted from the **Wikisource digitized text**
- New database part of the Trismegistos environment (de Graaf et al. 2024)
- <u>Unique identifier</u> for each entry

Extracting and modeling the information

Extracting information from Wikidata: enriching the database

- The case of M. Varro:
- **Instance of** → human
- **Citizenship** → Ancient Rome
- **Language** → Latin
- **Occupation(s)** → writer; poet; music theorist; philosopher; historian; grammarian; annalist; military personnel
- **Sex or gender** → male

Each mention is enriched with this information, while keeping it in its textual context.

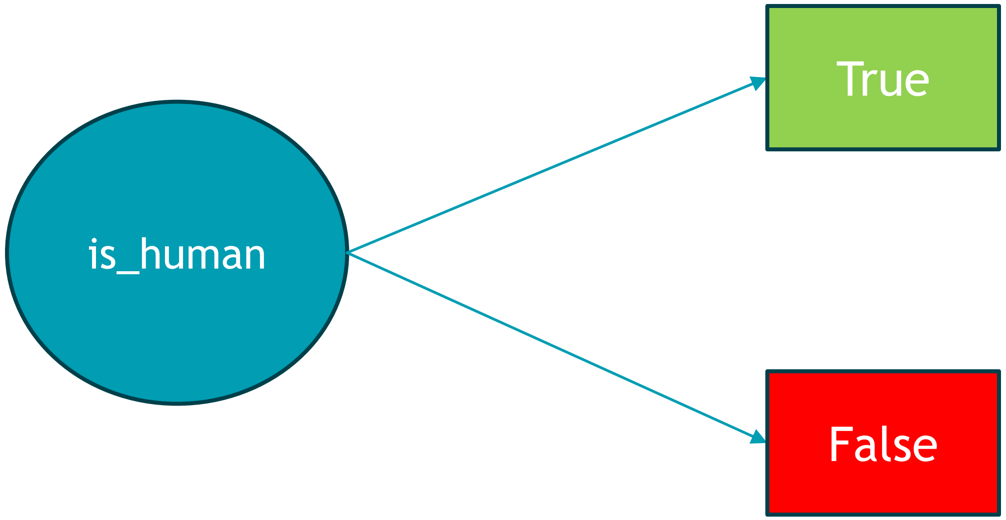
Extracting information from Wikidata: enriching the database

- The case of M. Varro:
- **Instance of** → human

The property values contain heterogenous information (e.g. citizenship: Ionian League, Miletus, Samos…):
we need to model the data in a uniform way

Each mention is enriched with this information, while keeping it in its textual context.

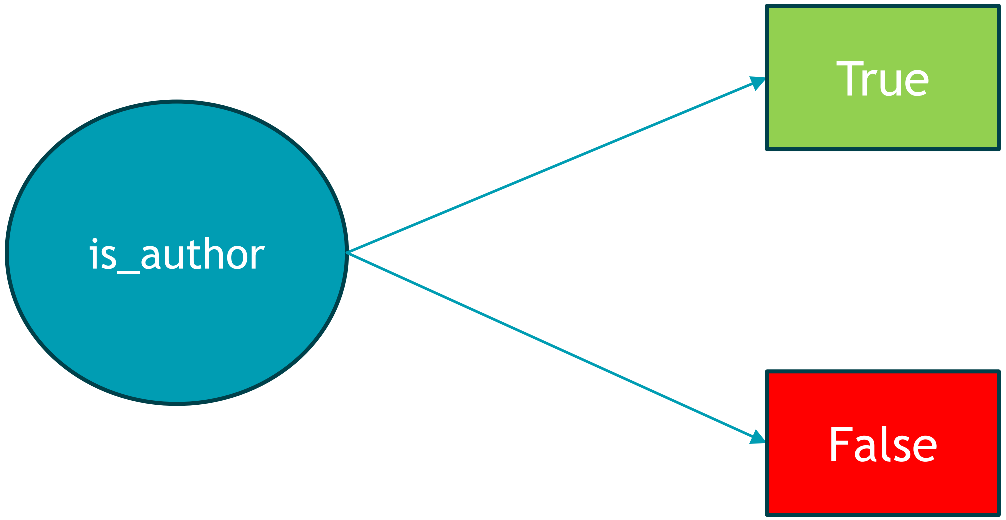
Extracting information from Wikidata: enriching the database



Based on the “instance of” (P31) value

- True: “human”, “human whose existence is disputed”, “prosopographical phantom”

Extracting information from Wikidata: enriching the database

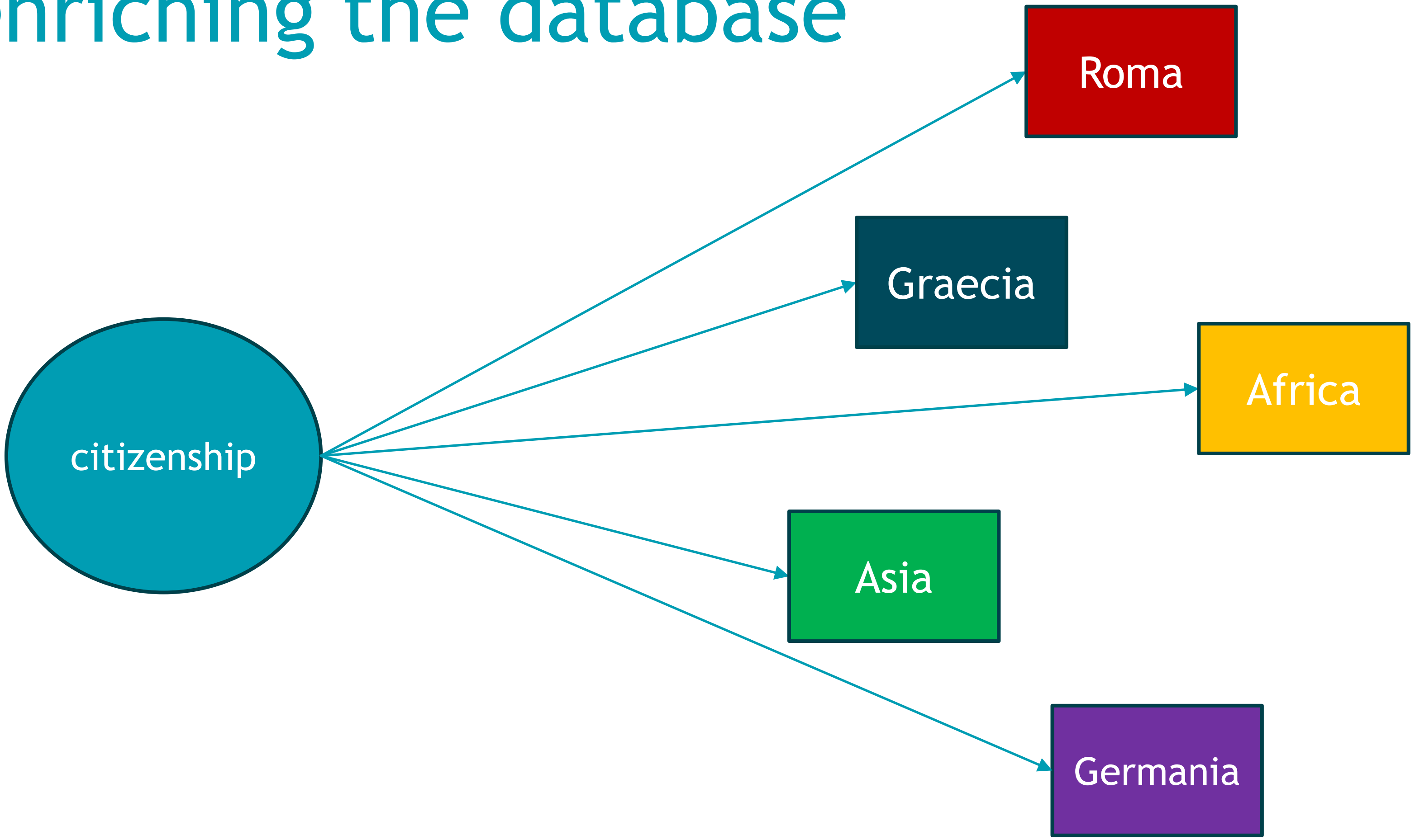


Based on the "occupation" (P106) value:

at least one "authorial" occupation in the list

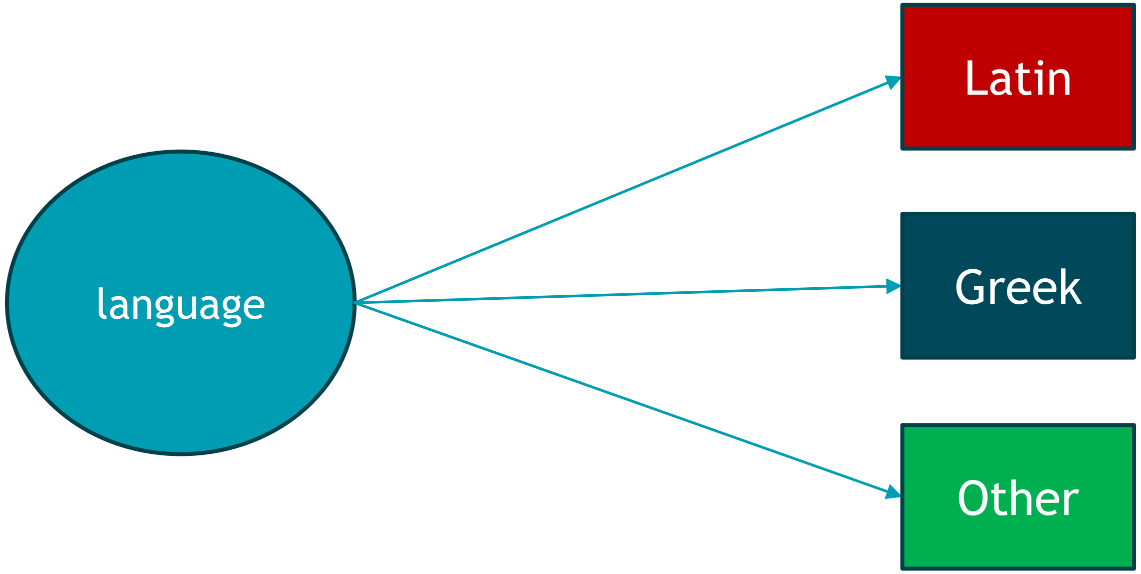
- Manual check of the results, to remove errors
- Comparison with list of sources in index, to spot any other inconsistency

Extracting information from Wikidata: enriching the database



Based on the “citizenship” (P27) value

Extracting information from Wikidata: enriching the database



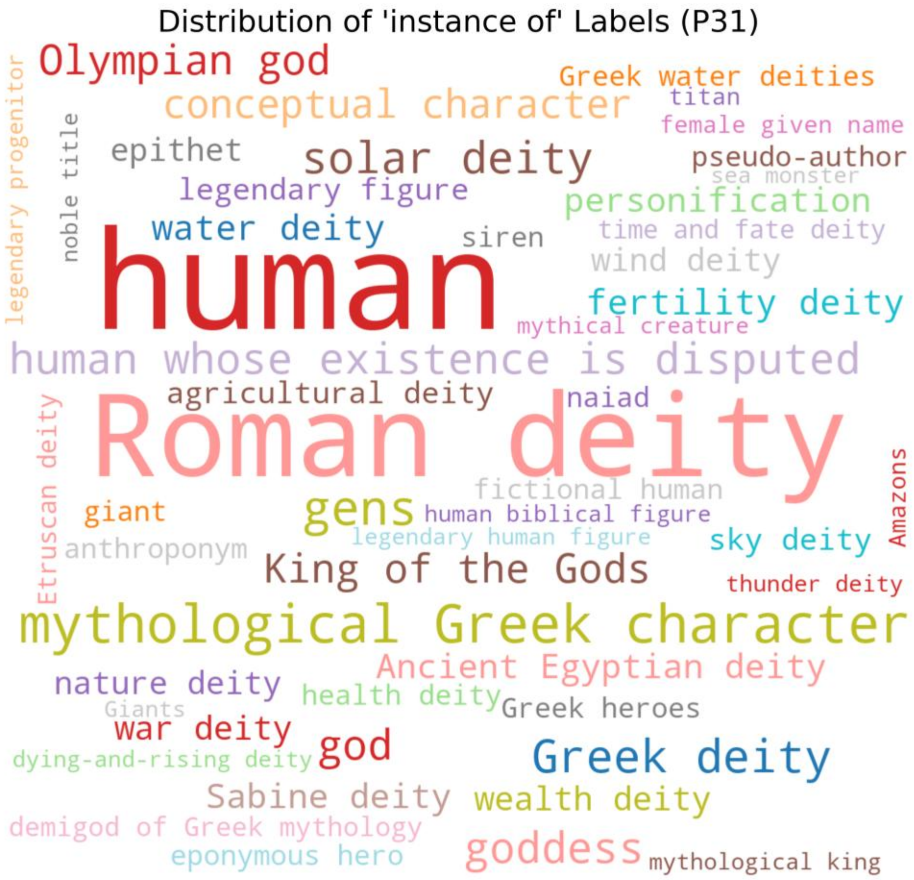
Based on the “languages spoken, written or signed” (P1412) value

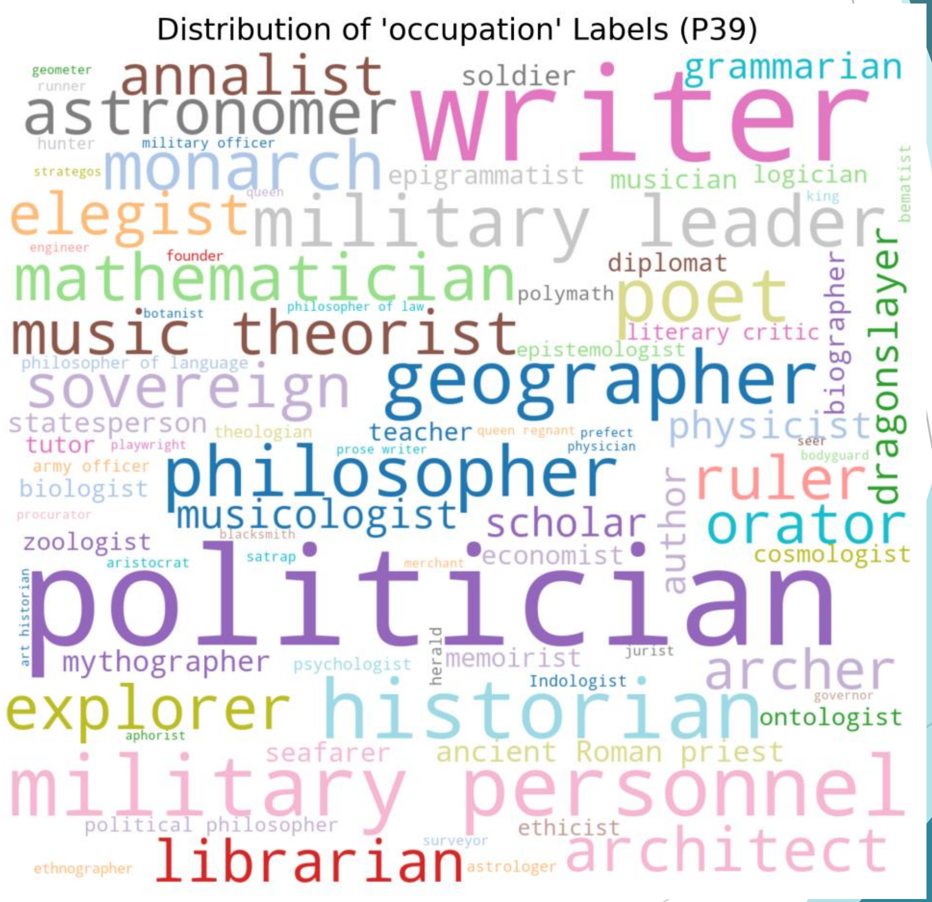
Extracting information from Wikidata: enriching the database

Geographical and cultural aspects do not align with our modern conception: e.g. many individuals of Greek culture and language lived on the African and Asian continent

Separate modelling of the two aspects allows to analyze each of them separately

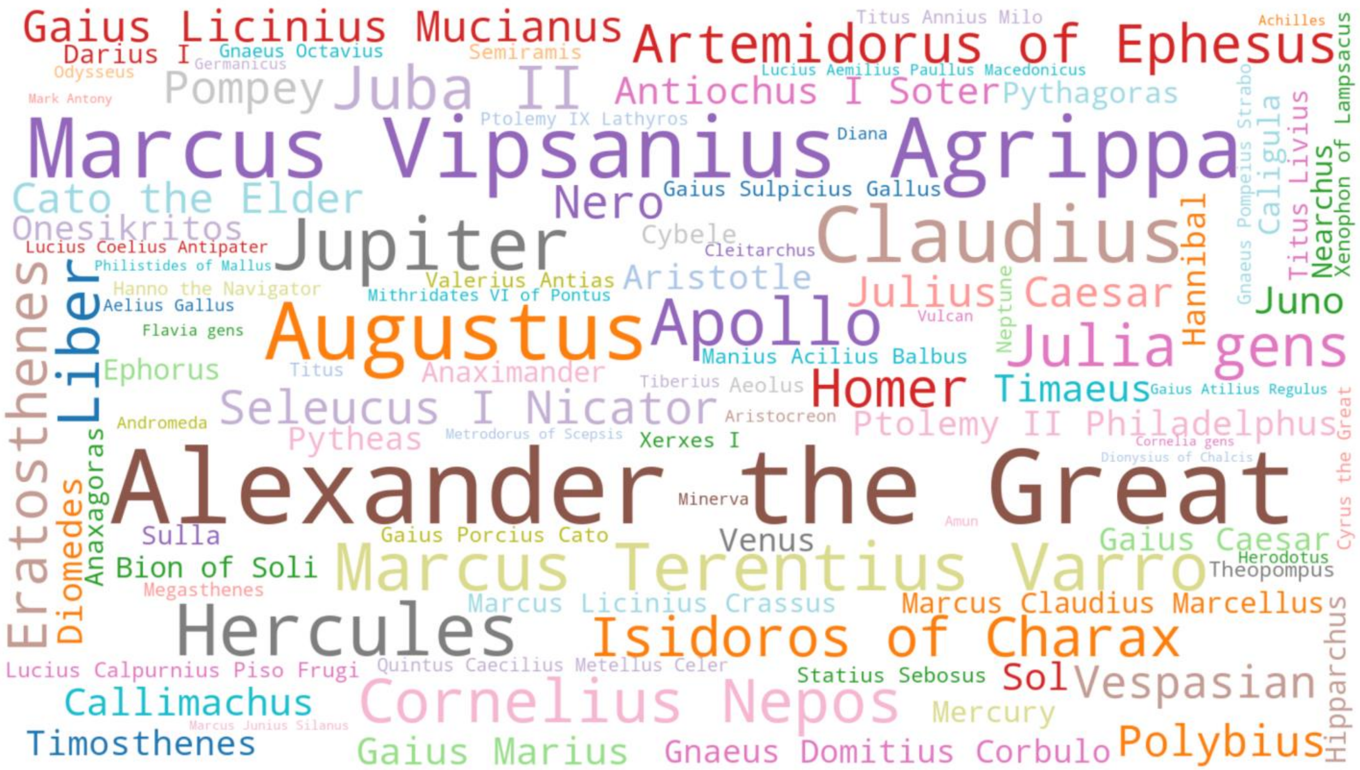
What kind of individuals?





Who? A look at the most cited Wikidata labels

Most cited individuals in book 2-6 of the Naturalis Historia: Wikidata labels



From annotation to interpretation

Pliny the Elder and the Greeks: admiration or contempt? A case study

Introduction : Pliny and the relationship with the Greeks

What is the attitude of Pliny towards the Greeks, the Romans and other nationalities mentioned in the books?

Scholarship is divided, specifically regarding the relation with the Greeks.

- **Critical relationship** with the Greeks, responsible of the moral decline of Rome (Carey 2003, pp. 23-25; Schilling 1978; French 1994).
- **Great admiration** of Greek authorities, and criticism only towards bad quality sources (Fögen 2013; Serbat 1987).

Greek science in general is characterized by misplaced ingenuity and vanity (Wallace-Hadrill 1990, p. 93).

Pline éprouve une admiration sans limite pour les savants grecs authentiques; mais une vive hostilité à l'égard de tous les charlatans hellénophones (Serbat 1986, pp. 2092–2093)

Introduction : Pliny and the relationship with the Greeks

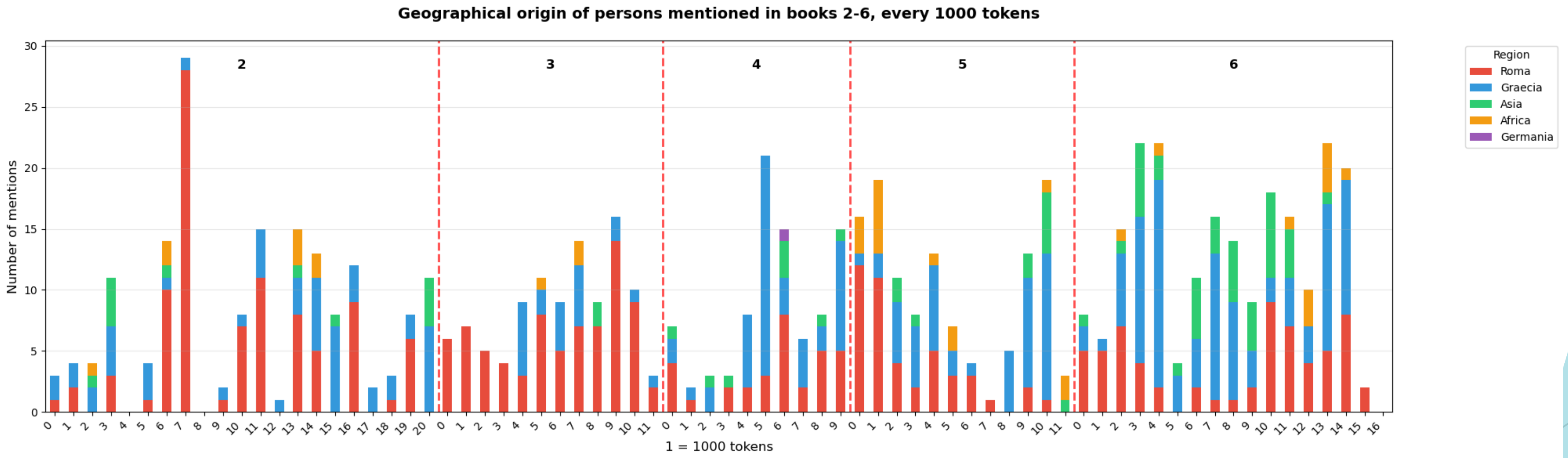
<u>Can semantic annotation of the text help us to shed a light on this problem?</u>

Using Python to perform a first general analysis of the distribution of different nationalities.

- *How many Greeks and Romans (and other geographical origins)?*
- *How often are they cited?*
- *How does this change when looking only at authors?*
- *What distribution in the text?*
- *How often are unique authors cited?*

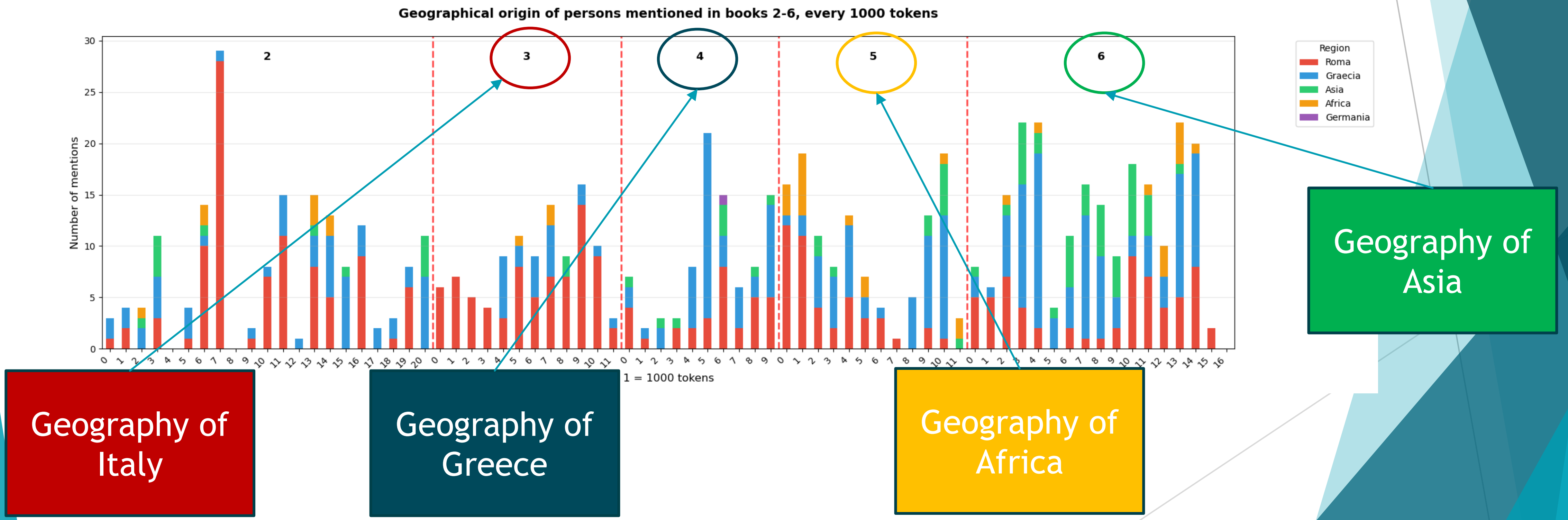
Distribution of different citizenships in the corpus

How many individuals of each nationality are mentioned every 1000 tokens?



Distribution of different citizenships in the corpus

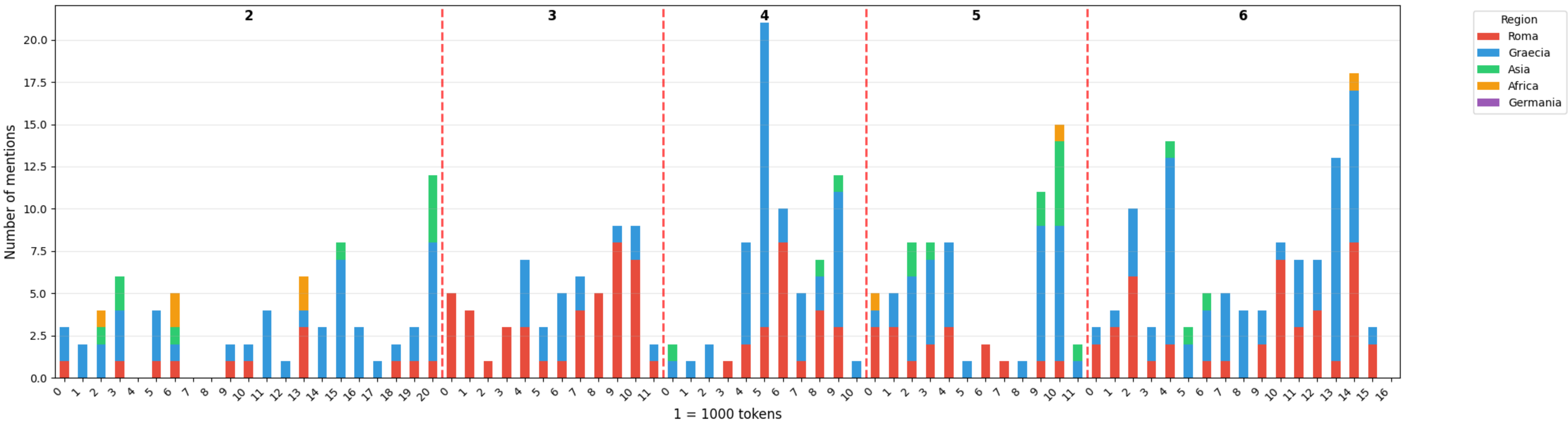
How many individuals of each nationality are mentioned every 1000 tokens?



What happens if we restrict the results only to (not) authors?

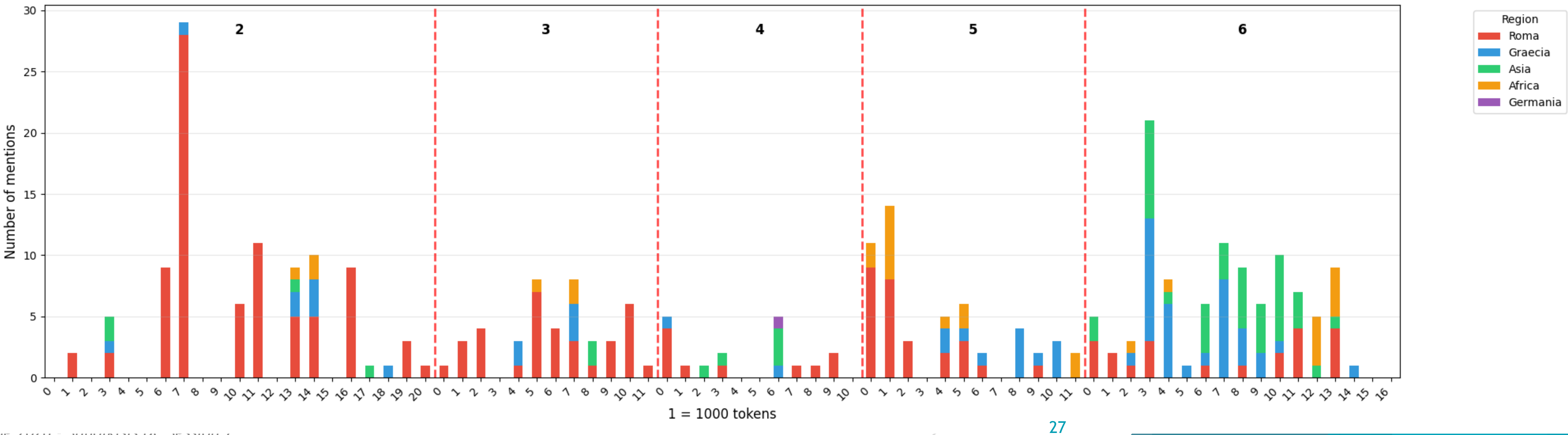
AUTHORS

Geographical origin of authors mentioned in books 2-6, every 1000 tokens



NOT AUTHORS

Geographical origin of not authors mentioned in books 2-6, every 1000 tokens



Distribution of different citizenships in the corpus: authors vs not authors

- If we restrict our results only to **authors**, we see that the Romans lose their predominance, in favor of the **Greeks**.

- **Not authors**, instead, are mostly **Roman citizens.**
  - Book 6 is an exception: here Alexander the Great (= Greek) is mentioned 32 times, and in 28 of those cases he is not mentioned as a source.

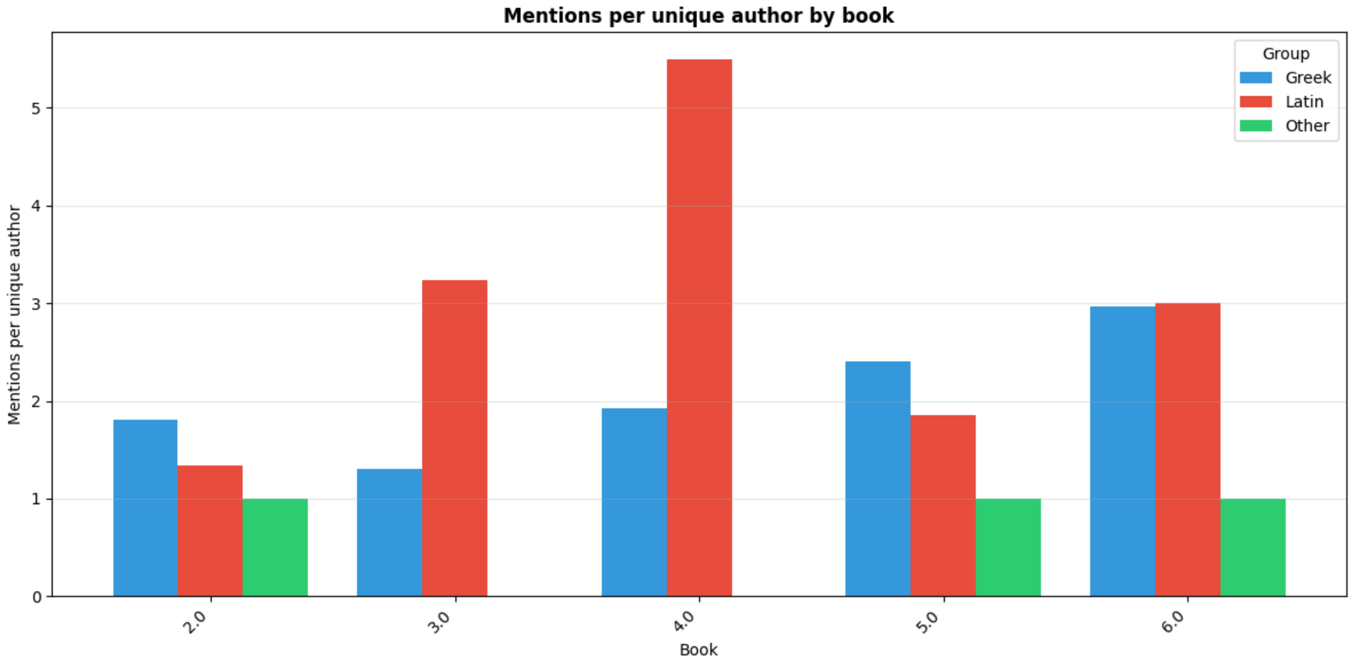
- Individuals pertaining to Africa and Asia mentioned in books 5 and 6 are mostly not authors

Now, let's compare the total mentions and the unique authors.

Focusing on authors: mentions vs unique individuals

We can inspect the data by looking at the ratio between the total mentions and the unique authors

= **How often is each unique author cited, on average?**



Less Latin authors, cited more often

Book 3

Language	Total mentions	Unique authors
Latin	42	13
Greek	17	13

Geography of Italy
Predictable (?)

Book 4

Language	Total mentions	Unique authors
Latin	22	4
Greek	48	25

Geography of Greece (!!!)

The Latin authors mentioned in book 4 are: Marcus Vipsanius Agrippa, Marcus Terentius Varro, Gaius Licinius Mucianus, and Cornelius Nepos.

- Agrippa alone is mentioned 12 times!

Less Latin authors, cited more often

- If we look at the totality of the 5 books, the difference is even more evident: **24 Latin authors are mentioned for a total of 116 times!**

Language	Total mentions	Unique authors
Greek	257	79
Latin	116	24
Other	5	3

This means that, on average:

- A Greek author is mentioned 3,25 times
- A Latin authors is mentioned 4,83 times

→ **Predominance of citations of Latin authors.**

First conclusions

- In terms of number of authors, Greeks are predominant → **majority of Greek sources.**
- However, **Latin sources are cited more often**: in particular, the top 5 authors are mentioned a great number of times.

- This gives us a first idea of **Pliny’s preference for Roman sources**, even if he could not do without the Greek ones.
- This does not automatically mean that he despised the Greeks.

Conclusion

Main take-aways

The use of semantic annotation can be fruitfully integrated within more traditional approaches to the study of text

- Importance of the **textual quality** when working with semantic information
- Integration of **automatic and manual approaches** to smooth the annotation process
- Importance of linking external information to elements of the text while keeping them in their **textual context**

Exercise

- You can find the exercise at this link (GoogleColab):
- https://colab.research.google.com/drive/1AlHM6tbCd1nWPRkhqre3CwCTWGYydBbb?usp=sharing
- **First create a copy on your own drive** (otherwise you will overwrite the code for everybody ☺)
- You will be able to perform the analysis on a restricted version of the corpus ("nh2_sample_sunoikisis.csv")
- Follow the instructions on the notebook!

Thank you for your attention!

Any questions?

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Ancient Greek Proper Names and the lead tablets from Styra

Digital Humanities for Greek and Latin Linguistics and Philology

9.6.2026

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Roadmap

1. The tablets from Styra (Euboea)
2. Between epigraphic and philological editing
3. The linguistics of proper names
4. Conclusions
5. Exercises

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1. The tablets from Styra (Euboea)

- **A corpus of c. 500 lead tablets** used as *tesserae publicae* (for voting, drawing lots and/or recording citizens' presences)

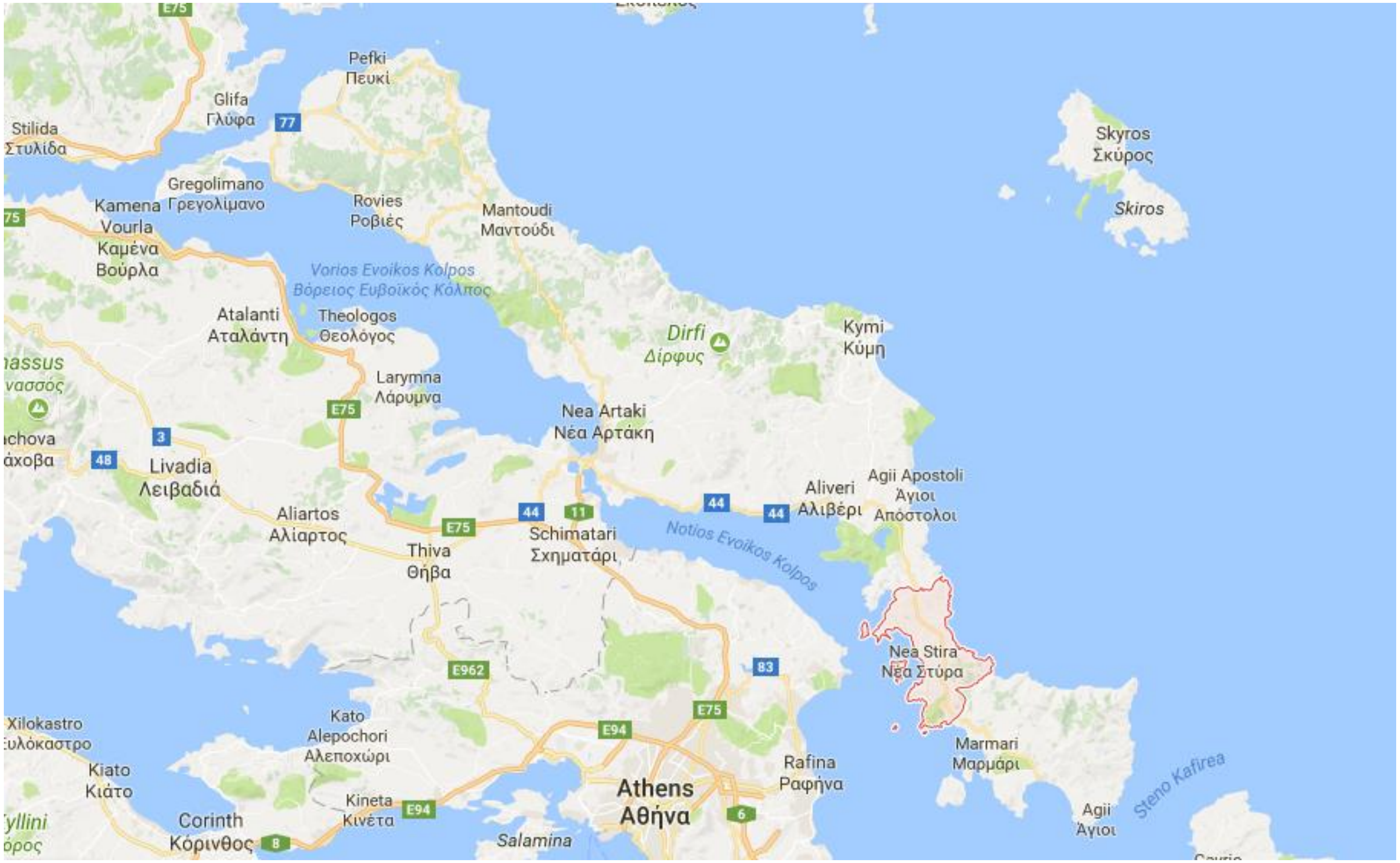
- **Dated to c. 475 BCE**

- **Discovered near Styra (Southern Euboea)**, deposited together after disposal; several tablets show clear evidence of reuse and multiple layers of writing

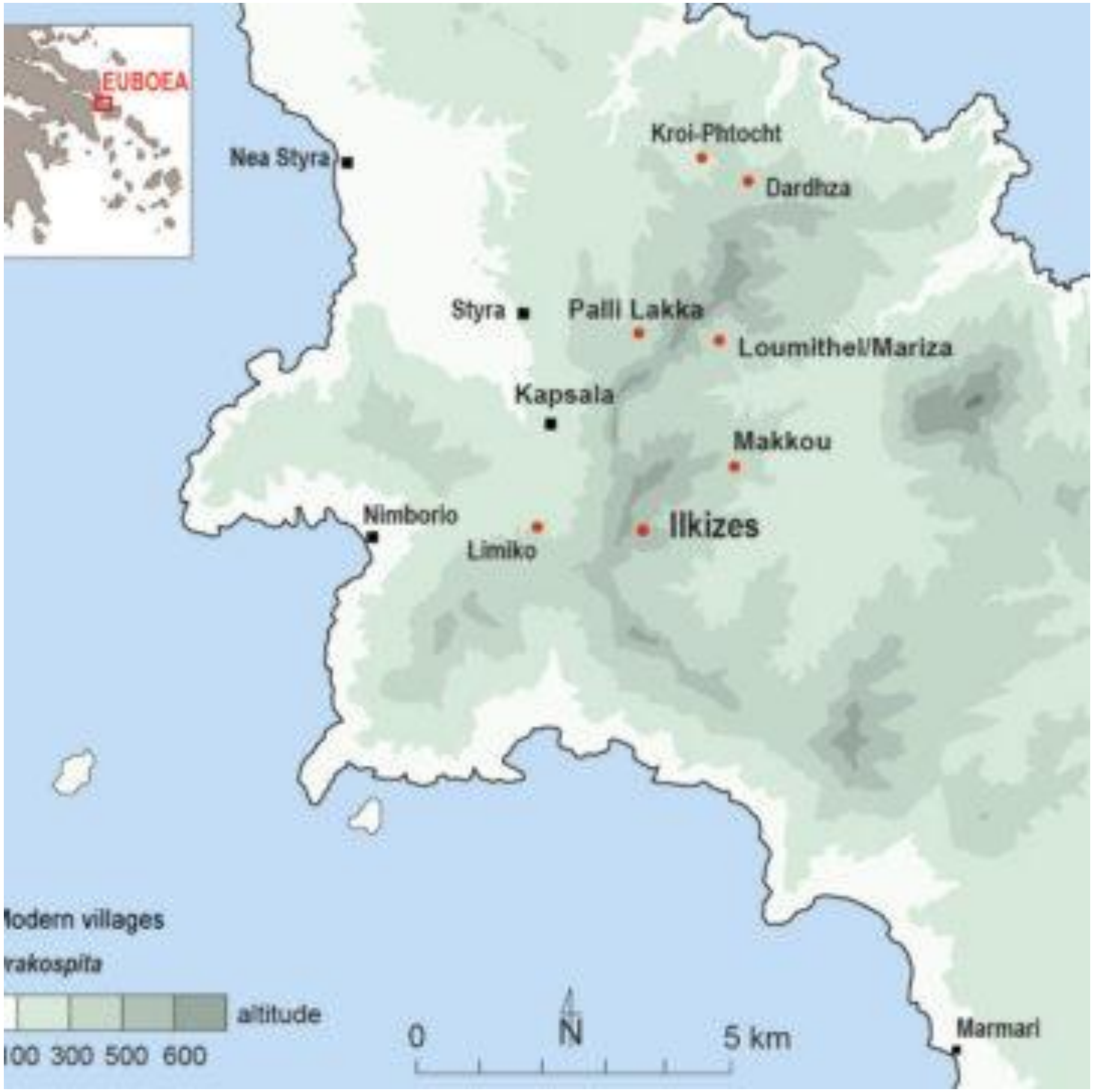
- **Mainly inscribed with proper names in the nominative**

- **Inscriptions in the Euboean dialect**, specifically a Southern Euboean variety

1. The tablets from Styra (Euboea)



1. The tablets from Styra (Euboea)



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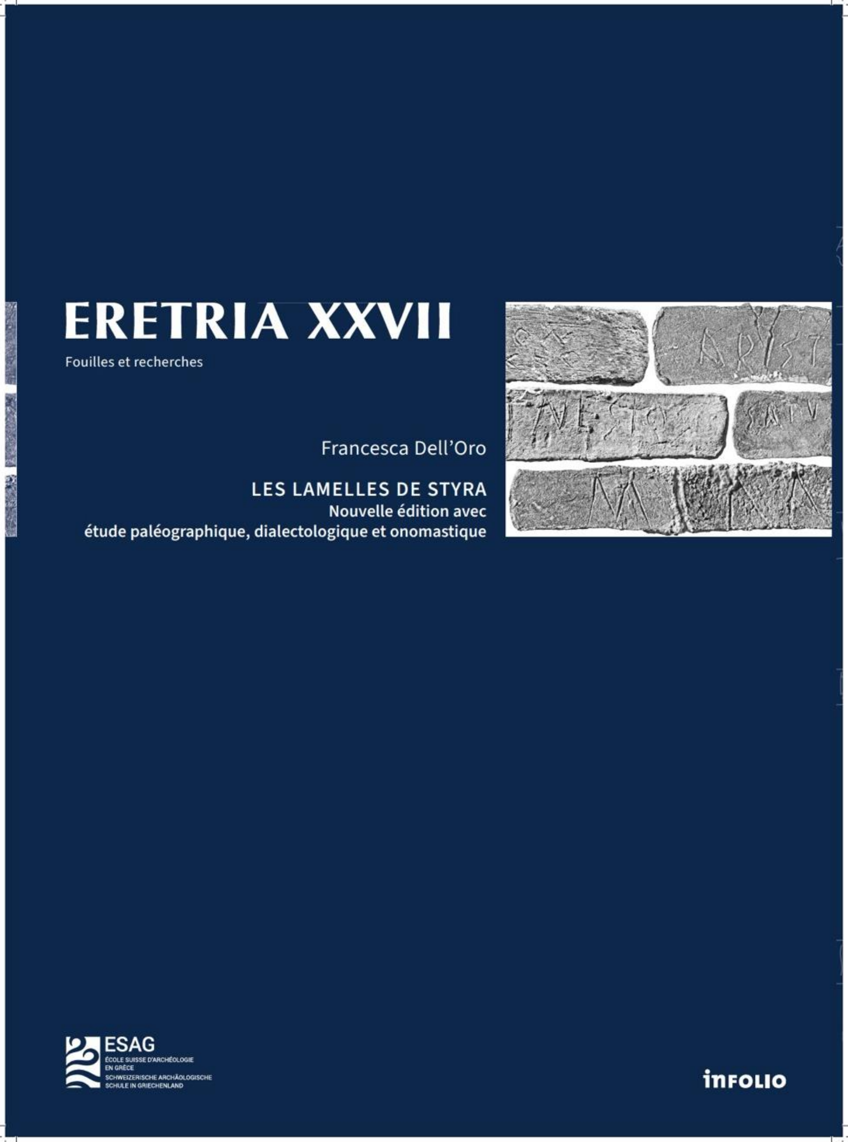

New edition of the tablets from Styra

https://www.esag.swiss/eretria-series/eretria-xxvii/

https://zenodo.org/records/17257282

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1. The tablets from Styra (Euboea)

- example of reuse

Α**ΝΤΙΜ**ΑΧΟ ΔΙ**ΚΙΟΣ** [ANTIMACHO DIKIOS]

2. Between epigraphical and philological edition

- **Epigraphical editorial approach**: extant and complete tablets

- **Philological editorial approach**: lost tablets known through earlier editions

- **Combined epigraphical-philological editorial approach**: fragmentary tablets

2. Between epigraphical and philological edition

- **Epigraphical editorial approach**: extant and complete tablets

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<text>
    <body>
        <div type="edition" xml:space="preserve">
        <ab>
            <lb n="1"/> <name nymRef="Master_names.xml#Αἰσχυλίων" ref="https://lgpn-ling.huma-num.fr/Aischyliōn">Αἰσχυλίο&#772;ν</name>
        </ab>
    </div>
```

2. Between epigraphical and philological edition

- **Philological editorial approach**: lost tablets known through earlier editions

```
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        <div type="edition" xml:space="preserve">
        <ab>
            <lb n="1"/>
            <app>
<lem>
   <name nymRef="Master_names.xml#Ἀγλωχάρης"
         ref="https://lgpn-ling.huma-num.fr/Aglōchares">Ἀγλοχάρε&#772;ς</name>
</lem>
<rdg resp="#Korte">Ἀγλωχάρης</rdg>
<rdg resp="#Vischer">Λόχαγος</rdg>
</app>
        </ab>
    </div>
```

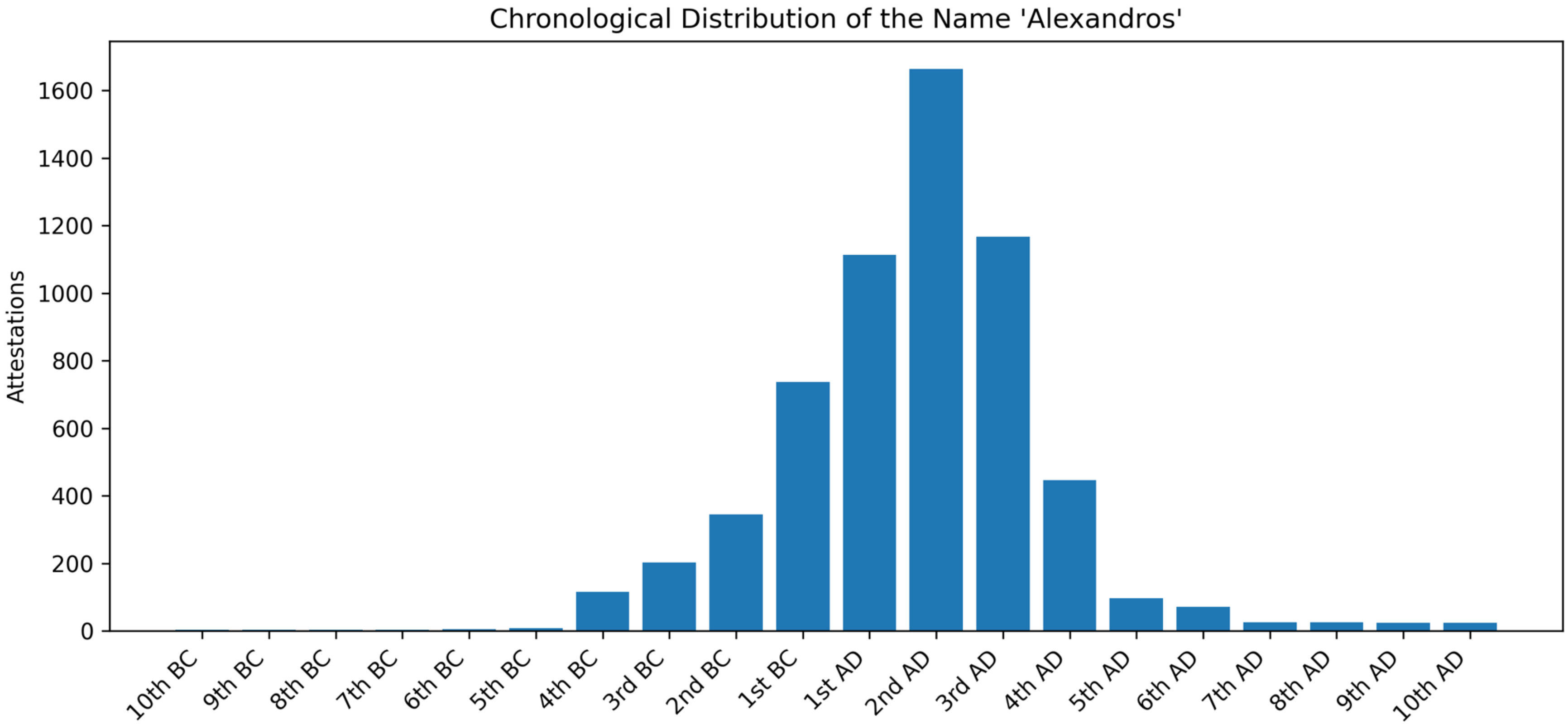
3. The linguistics of proper names

- Semantic transparency was a pervasive feature of personal names in Ancient Greece
    - *Demokritos* 'chosen by the people'
    - *Perikles* 'very glorious'
    - *Sokrates* 'of safe strength'

- This is no longer the case in modern English or Italian (but see *Joy* or *Gioia*)

- Psychological and socio-cultural factors
    - *Alexandros* 'defending the men' → after the conquests of Alexander the Great (356–323 BCE), the name became closely associated with this historical figure

3. The linguistics of proper names



Data taken from the LGPN database

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3. The linguistics of proper names

LGPN: historical/geographical distribution
LGPN-Ling: linguistic analysis (morphology and semantics)

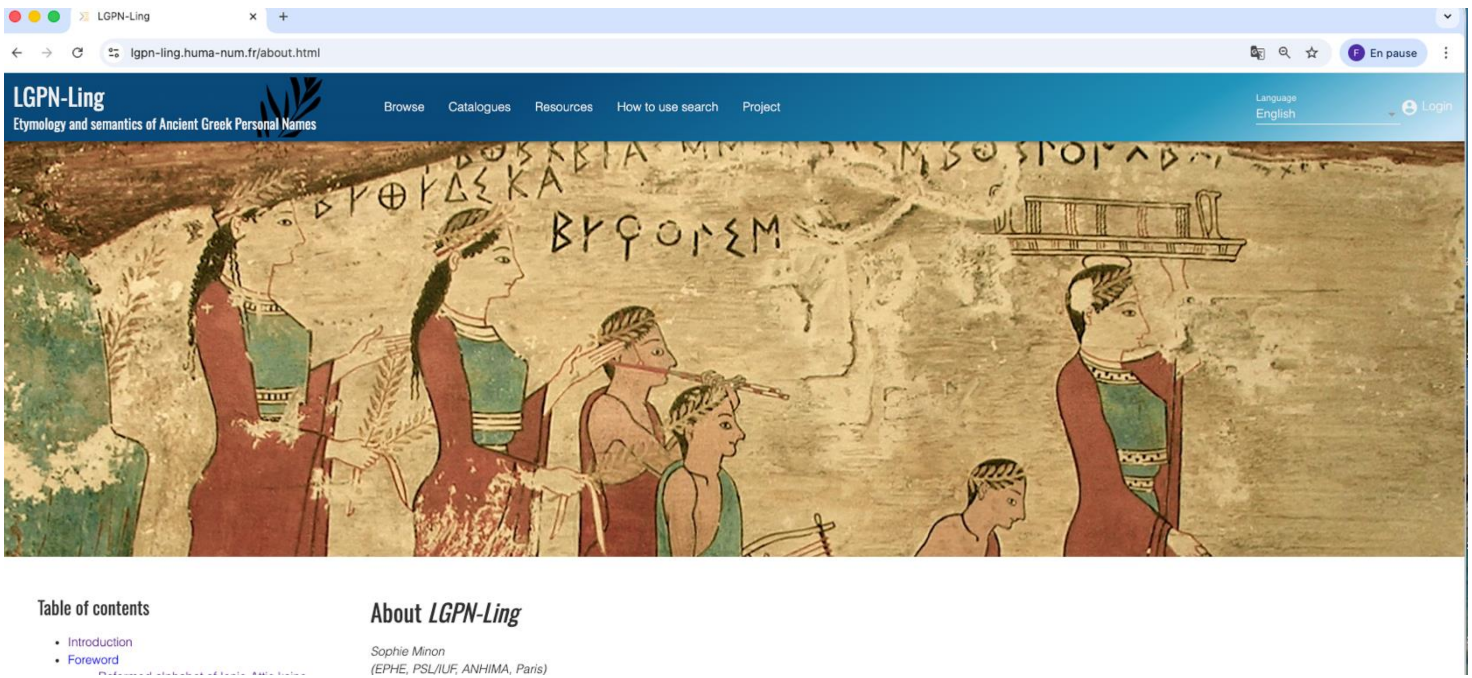
If you want to know more about the geographical and historical distribution of Ancient Greek personal names:

https://www.lgpn.ox.ac.uk/

3. The linguistics of proper names

LGPN: historical/geographical distribution
LGPN-Ling: linguistic analysis (morphology and semantics)

If you want to discover the meanings of Ancient Greek personal names:
https://lgpn-ling.huma-num.fr/about.html



3. The linguistics of proper names

|< 1 >| Found 1 item

Sort by: Name | Filter by: Name | Filter: aglochares

Name	Geolinguistic		Bases		PN Suffixes				Bases Morphosyntax (PN Genetics)	Semantics
		Prefix	Root1	Root2	S.4	S.3	S.2	S.1		
Ἀγλωχάρης Aglōcharēs *-ης* Early *-εος*, Att. *-ους*, A.M. *-ευς*, then *-ε(ι)ου(ς)* LGPN 2[m.] -500/-400 Trismegistos	Euboean, Cycladic Ionic		**αγλα(ϝ)(ο)**	**χαρ-(εσ)**					Adj—V YX	sparkling, brilliant-rejoiced
Ἀγλώχαρος	Cycladic Ionic	m								

3. The linguistics of proper names

- to avoid repetition, in the digital edition of the tablets, reference is made to the LGPN-Ling database:

```
<name nymRef="Master_names.xml#Ἀγλωχάρης"
      ref="https://lgpn-ling.huma-num.fr/Aglōcharēs">Ἀγλοχάρε&#772;ς</name>
```

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3. The linguistics of proper names

- Editions may enrich the text with additional linguistic and onomastic metadata

- For example: dialectal variants, frequency of attestation (hapax), external identifiers

- Such information can be managed through an authority file

3. The linguistics of proper names

```xml
<nym xml:id="Τιμίνης">
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        <usg type="dialect">Euboean</usg>
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    </form>
    <form type="variant">
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        <usg type="dialect">North-West Doric</usg>
        <usg type="dialect">Boeotian</usg>
        <ref type="LGPN-Ling">https://lgpn-ling.huma-num.fr/Timinas</ref>
    </form>
    <note type="editorial">
        LGPN records the name as
        <ref target="http://clas-lgpn2.classics.ox.ac.uk/Τιμνίης/">Τιμνίης</ref>,
        following an earlier reading rejected in the present edition.
    </note>
</nym>
```

4. Conclusions

This presentation has shown

- how epigraphic and philological approaches were combined both in editing the tablets from Styra and in encoding their edition using XML-TEI

- how the linguistic features of Ancient Greek personal names can be identified, analysed and encoded using XML-TEI

EXERCISE 1

1. Using the TEI Guidelines and the example discussed above, suggest an encoding for the apparatus the following tablet from Styra. The tablet is no longer extant, and earlier editors have proposed different readings.

STY 263 = LOU 48

Identification : Lambros 42 = MNB 3188.63. Conservation : lamelle introuvable. Éditions : liste Vischer – Lambros nº 113 (liste Lambros nº 42), Lenormant (1867 : nº 53), Roehl (1882 : nº 112), Bechtel (1887 : nº 197), Audollent (1904 : nº 120), Ziebarth (1915 : nº126). Dessins : Lenormant (1867 : nº 53)[Roehl (1882 : nº 112)].

Εὐτέλε̄ς

Apparat : ΕΥΤΕΛΕΣ Vischer, Εὐτέλης Bechtel : Εὐτελές Audollent, Εὐγένης Lenormant.

```
…
<div>
          <div type="edition" xml:space="preserve">
   <ab>
      <lb n="1"/>
      <app>
         <lem> … </lem>
         <rdg resp=  "…">. … </rdg>
      </app>
   </ab>
</div>
```

Solution exercise 1

```
<div type="edition" xml:space="preserve">
    <ab>
        <lb n="1"/>
        <app>
            <lem>
                <name nymRef="Master_names.xml#Εὐτέλης"
                      ref="https://lgpn-ling.huma-num.fr/Eutelēs">Εὐτέλες</name>
            </lem>
            <rdg resp="#Vischer">ΕΥΤΕΛΕΣ</rdg>
            <rdg resp="#Bechtel">Εὐτέλης</rdg>
            <rdg resp="#Audollent">Εὐτελές</rdg>
            <rdg resp="#Lenormant">Εὐγένης</rdg>
        </app>
    </ab>
</div>
```

EXERCISE 2

2) encode the diatopic variants of the name Κλεογενίδης/Κλογενίδης (STY 180) in the authority file

Name	Geolinguistic	Bases			PN Suffixes				Bases Morphosyntax (PN Genetics)	Semantics
		Prefix	Root1	Root2	S.4	S.3	S.2	S.1		
Κλευγενίδας Kleugenidas *-ᾱ/ης* Att. *-ου*, Ion., A.M. *-εω* etc., Beot. *-ᾱο*, Dor. *-ᾱ* LGPN 1[m.] -600/-500 Trismegistos	South Doric		**κλε(ϝ)-(ε)(σ)**	**γεν-(εσ)**			(ι)δ	*ᾱ/ης*	N—N YX	fame, glory-kin, birth, origin, born
Κλεογενίδης	Euboean	m								
Κλογενίδης	Euboean	m								

<nym xml:id="Κλεογενίδης">

<form type="lemma">
<orth> … </orth>
<usg type="dialect"> … </usg>
</form>

<ref type="LGPN-Ling"> … </ref>

</nym>

Solution exercise 2

```xml
<nym xml:id="Κλεογενίδης">

    <form type="lemma">
        <orth>Κλεογενίδης</orth>
        <usg type="dialect">Euboean</usg>
    </form>

    <form type="variant">
        <orth>Κλογενίδης</orth>
        <usg type="dialect">Euboean</usg>
    </form>

    <form type="variant">
        <orth>Κλεογενίδας</orth>
        <usg type="dialect">South Doric</usg>
    </form>

    <ref type="LGPN-Ling">https://lgpn-ling.huma-num.fr/Kleugenidas</ref>

</nym>
```

ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Thank you!

Merci!

Grazie!

francesca.delloro@unibo.it
delloro.fr@gmail.com

References

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LGPN-Ling: https://lgpn-ling.huma-num.fr/about.html

Dell'Oro, F. 2026, forthcoming. Linguistics and Inscriptions. In: Elton Barker, Olympia Bobou e Rubina Raja, eds, *Oxford Handbook of Digital Classical Studies*, Oxford: Oxford University Press.

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Masson, O. 1992. Les lamelles de plomb de Styra, IG XII 9,56: essai de bilan. *Bulletin de Correspondance Hellénique* 116, 61–72. https://www.persee.fr/doc/bch_0007-4217_1992_num_116_1_1695

SunoikisisDC Summer 2026

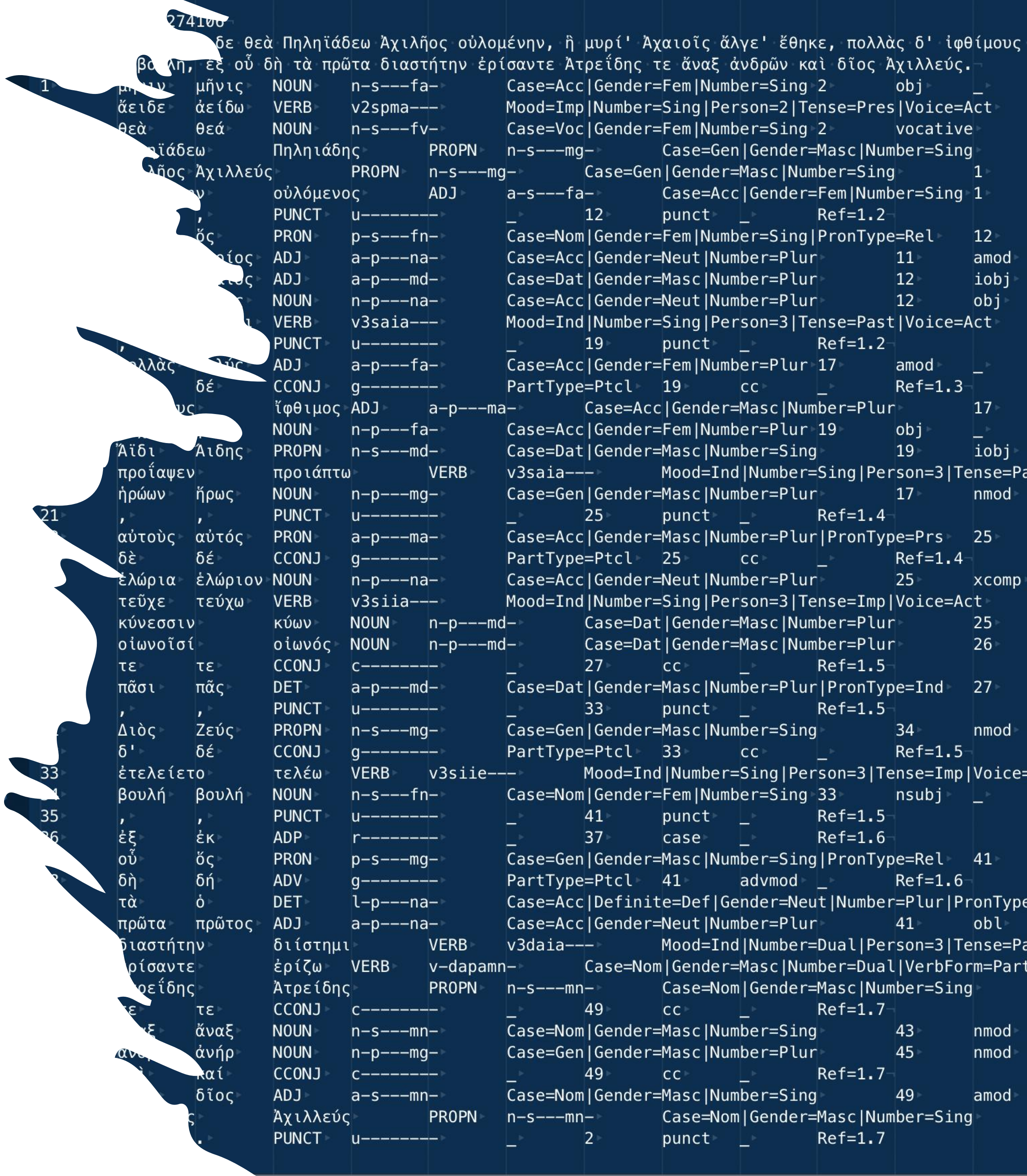
Session 7, 9 June 2026

Digital Humanities for Greek and Latin Linguistics and Philology

Part III

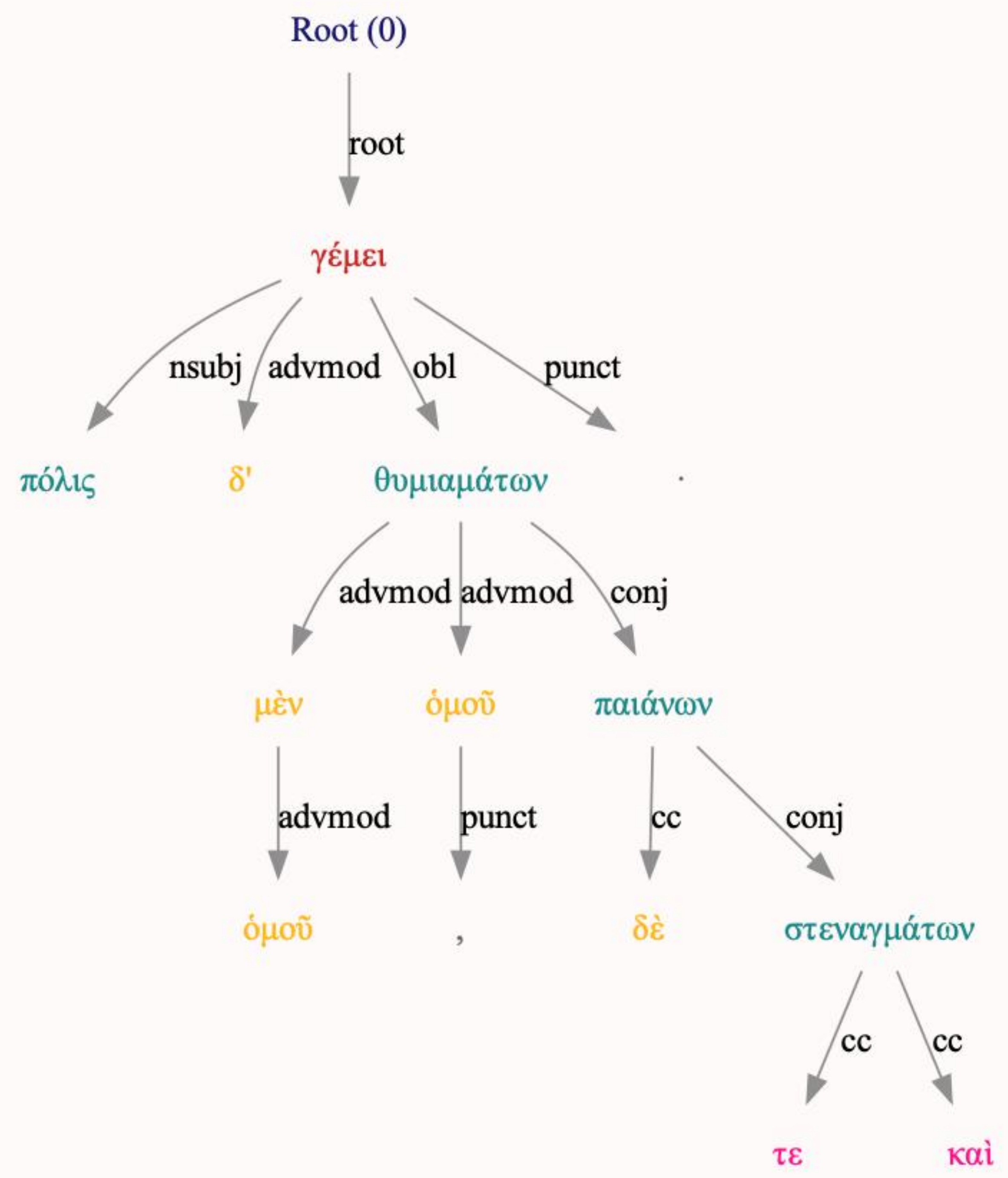
Francesco Mambrini

Università Cattolica del Sacro Cuore Milano



Treebank

- **Annotated corpus**: lemmatization, POS, morphological features, syntax
- Text is **divided into sentences**; every **«token»** in every sentence is annotated
- Treebanks of Ancient Greek (and Latin) adopt a **formalism** for syntactic annotation based on (different forms of) **dependency grammar**
- Some of these also provided converted versions of their treebanks to a project called UD



A (sort of) timeline

2007
ITTB and LDT (Perseus)

2008
PROIEL treebank in Oslo

2009
AGLDT (Perseus): 200k annotated tokens

2014
Universal Dependencies 1.0, AGLDT 2.0

2018
Gorman Trees, Pedalion, Sematia/PapyGreek

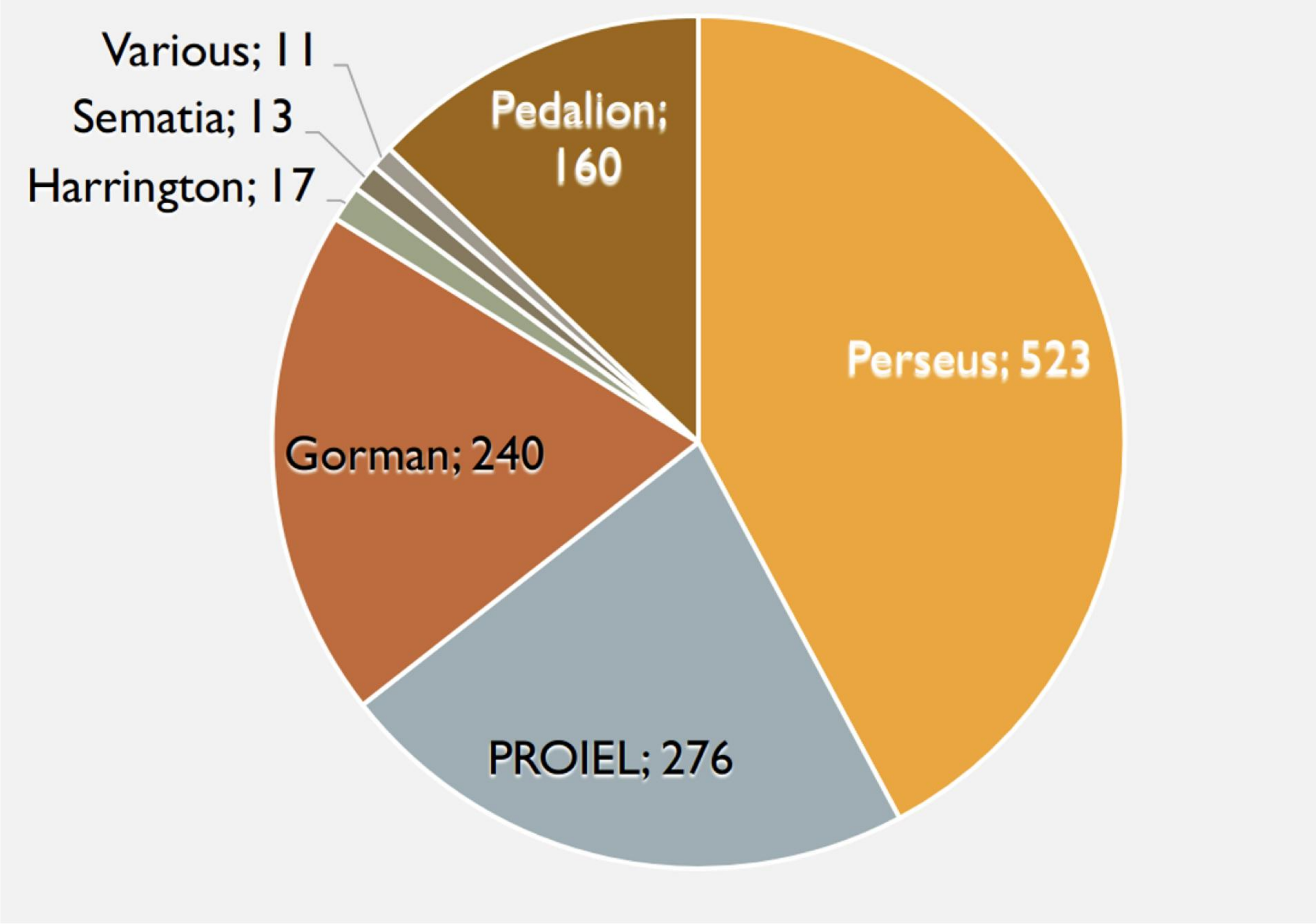
2022
Glaux (Alek Keersmaekers)

2024
Opera Graeca Adnotata (Celano)

Now
We went a very long way 🥳!

AG Treebanks: the “actual” (2019) situation

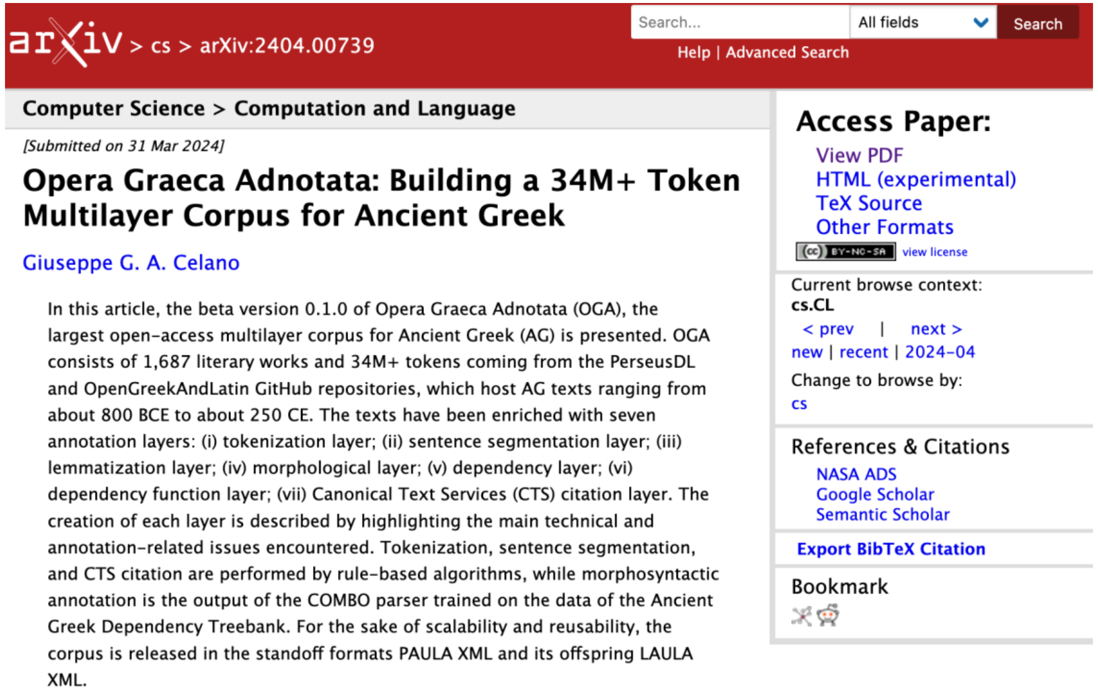
- **AGLDT**: poetry + prose (V. Gorman); mostly complete works.
- **Gorman**: many prose texts (mostly not in AGLDT).
- **PROIEL**: multilingual tb with NT e un selection of Hdt.
- **Pedalion**: from Leuven! Annotation to support a digital grammar of AG.
- And currently (2025), **much more!**

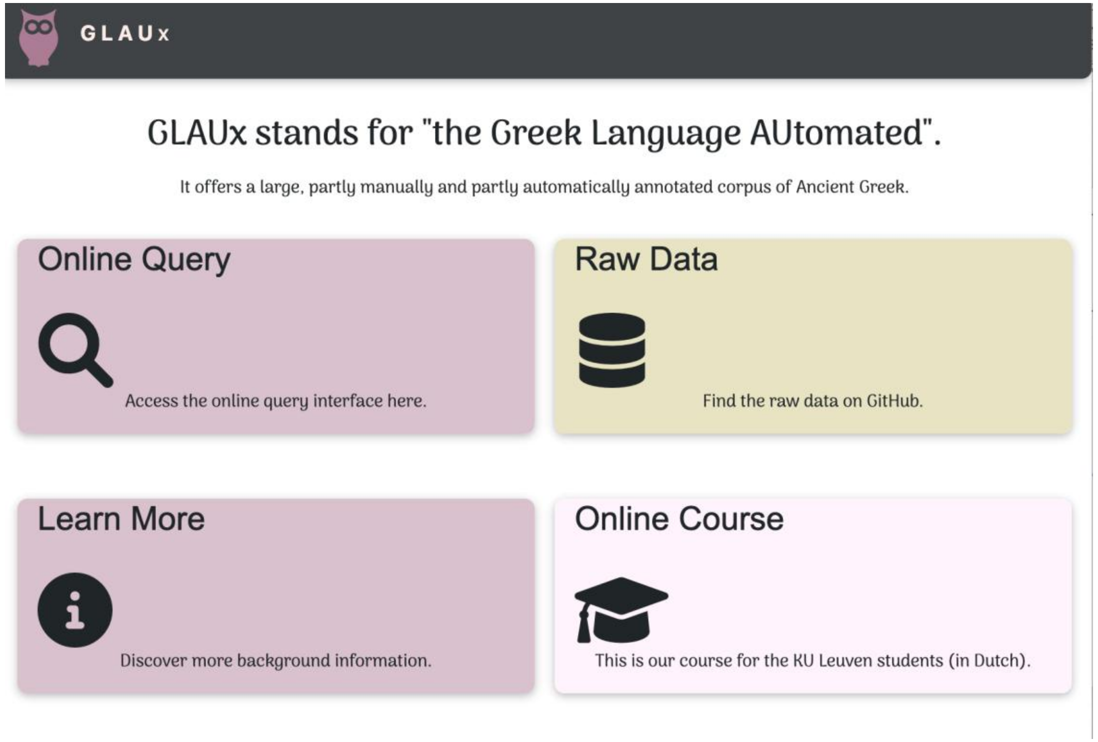


Keersmaekers et al. 2019

https://syntaxfest.github.io/syntaxfest19/slides/slides_68.pdf

Automatic annotation bonanza!





Opera Graeca Adnotata (34M+ token)

https://doi.org/10.5281/zenodo.14206061

Glaux (20M+ token)

https://github.com/alekkeersmaekers/glaux

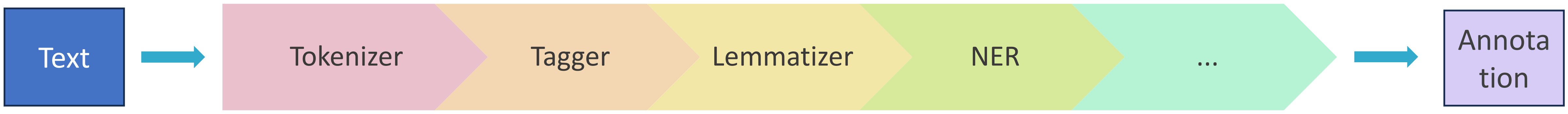
Morphology


```xml
<sentence subdoc="4-5" id="2386547" document_id="urn:cts:greekLit:tlg0011.tlg004.pe
  <annotator>FrancescoM</annotator>
  <word id="1" form="πόλις" lemma="πόλις" postag="n-s---fn-" head="6" relation="SBJ
  <word id="2" form="δ’" lemma="δέ" postag="g--------" head="6" relation="AuxY" cit
  <word id="3" form="ὁμοῦ" lemma="ὁμοῦ" postag="d--------" head="4" relation="AuxZ"
  <word id="4" form="μὲν" lemma="μέν" postag="g--------" head="9" relation="AuxY" c
  <word id="5" form="θυμιαμάτων" lemma="θυμίαμα" postag="n-p---ng-" head="9" relati
  <word id="6" form="γέμει" lemma="γέμω" postag="v3spia---" head="0" relation="PRED
  <word id="7" form="," lemma="," postag="u--------" head="9" relation="AuxX" cite=
  <word id="8" form="ὁμοῦ" lemma="ὁμοῦ" postag="d--------" head="9" relation="AuxZ"
  <word id="9" form="δὲ" lemma="δέ" postag="g--------" head="6" relation="COORD" ci
  <word id="10" form="παιάνων" lemma="Παιάν" postag="n-p---mg-" head="12" relation=
  <word id="11" form="τε" lemma="τε" postag="g--------" head="12" relation="AuxY" c
  <word id="12" form="καὶ" lemma="καί" postag="c--------" head="9" relation="COORD"
  <word id="13" form="στεναγμάτων" lemma="στέναγμα" postag="n-p---ng-" head="12" re
  <word id="14" form="·" lemma="·" postag="u--------" head="0" relation="AuxK" cite
</sentence>
```


TEXT

Syntax

Annotation via pipeline



The annotation is typically produced via a **pipeline**, i.e. a chain of modules where the output of a service becomes the input of the following one. To produce a treebank like those that we normally use, we start with **tokenization** (i.e. segmenting the text into sentences and "words"), then we add **POS tagging** , (tagging of the morphology), **lemmatizaion** and **syntactic parsing.**

How is the situation in the Greek treebanks now?

Fine, except for inconsistencies in…

1. Character encoding
2. Tokenization
3. POS tagging
4. Lemmatization
5. Use of UD’s morphological features
6. Syntax

Limits of the status quo

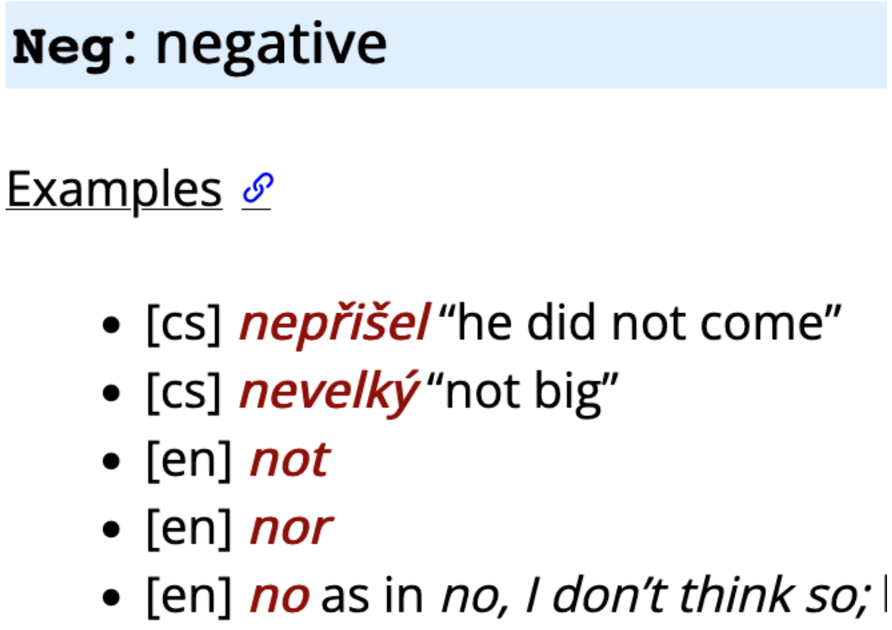
- I don't like having just pre-annotated data (what if my text is not there?) → **models with data**, please!
- What if I don't like the **format**?
- The AGLDT format is not great for typologically-oriented research (it's used for Greek and Lat. only)
- What about **research tools**?
- What about **extensions** (other layers)?
- **Accuracy** is not great! (OGA reports 63.55% LAS)
- There is **room for improvement!**

Universal Dependencies (UD)

- What:
  - A community standard
  - A set of guidelines for morphosyntactic annotation
  - A collection of treebanks
- Why:
  - It is supported by a large community
  - Many tools are already available
  - Many extensions
  - Promotes interlinguistic and typological research and interoperability between syntactic annotation

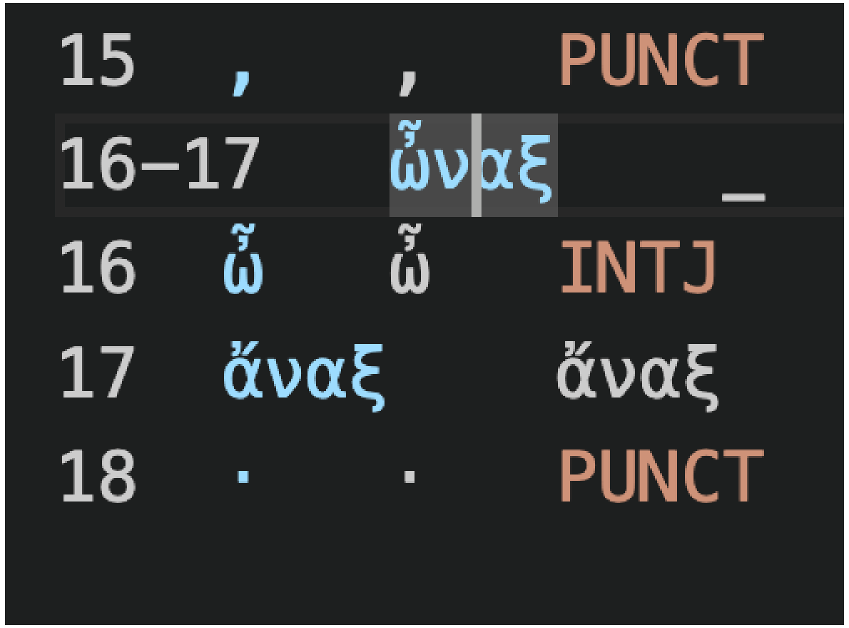
https://universaldependencies.org/

In UD we can get rid of many sources of variation


Neg: negative

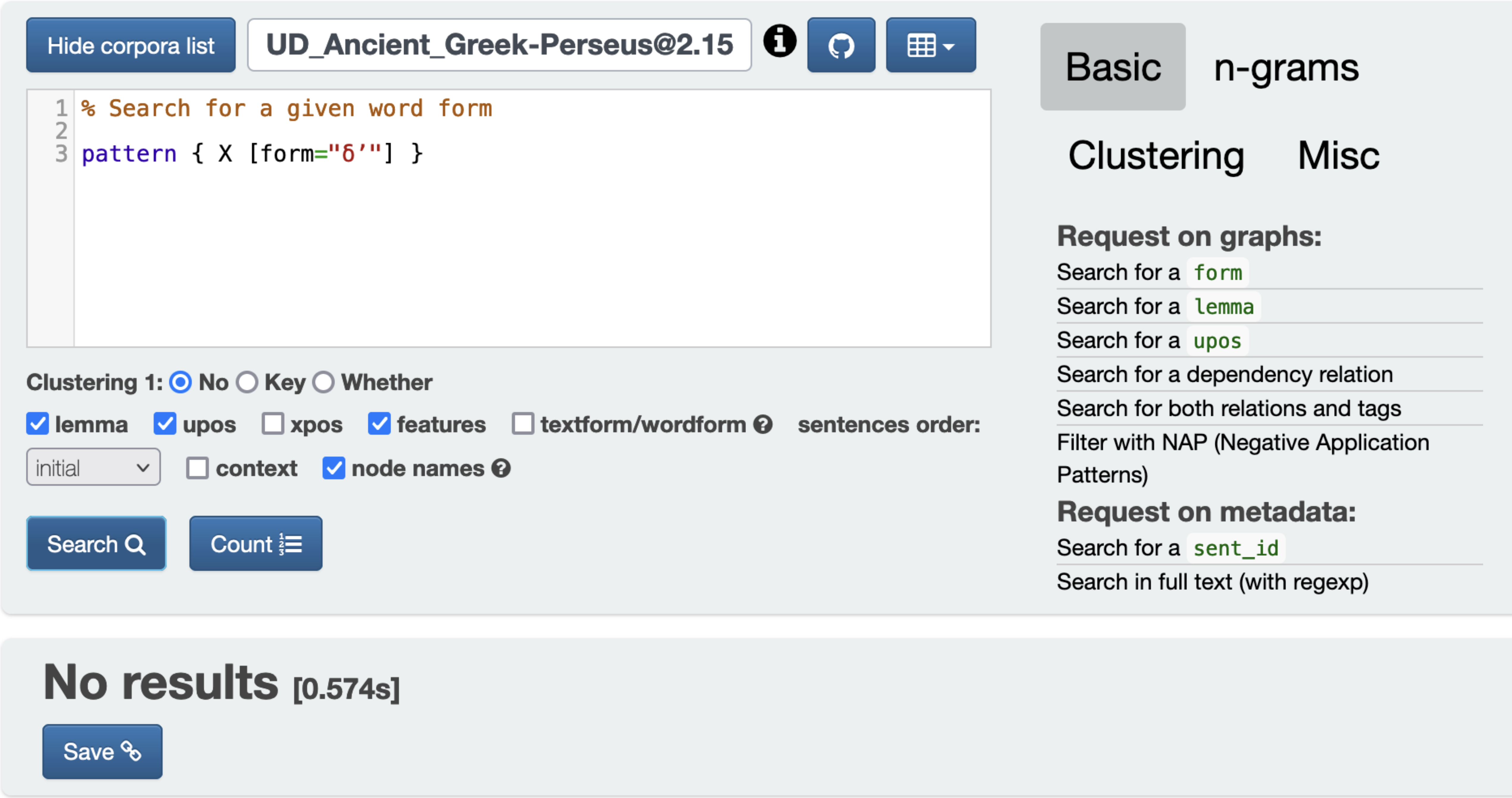
Examples

- [cs] *nepřišel* “he did not come”
- [cs] *nevelký* “not big”
- [en] *not*
- [en] *nor*
- [en] *no* as in *no, I don’t think so;*



Because conjunction don’t play a structural role in the trees (they are leaf node); and we can express negative polarity as a morph feature!

And because UD provides a formalism to express contracted words as “multiword tokens”, thus keeping the text and expanding the components


Hide corpora list
UD_Ancient_Greek-Perseus@2.15
Basic
n-grams
Clustering
Misc
1 % Search for a given word form
2
3 pattern { X [form="δ’"] }
Clustering 1: No Key Whether
lemma upos xpos features textform/wordform sentences order:
initial context node names
Search
Count
Request on graphs:
Search for a form
Search for a lemma
Search for a upos
Search for a dependency relation
Search for both relations and tags
Filter with NAP (Negative Application Patterns)
Request on metadata:
Search for a sent_id
Search in full text (with regexp)
No results [0.574s]
Save

Error detection in dependency arcs

1. Rule-based approaches (UD validation; Dickinson 2015)
2. Context-based approaches: n-gram variation (Meurers and Dickinson 2005); variation in nuclei (Boyd et al. 2008, de Marneffe et al. 2017)
3. Unsupervised rating of arcs (LISCA: Dell'Orletta et al. 2013, Alzetta et al. 2018…)

“MarkBugs” in Udapi

```python
class MarkBugs(Block):
    """Block for checking suspicious/wrong constructions in UD v2."""

    def __init__(self, save_stats=True, tests=None, skip=None, max_cop_lemmas=2, **kwargs):
        """Create the MarkBugs block object.

        Args:
        save_stats: store the bug statistics overview into `document.misc["bugs"]`?
        tests: a regex of tests to include.
            If `not re.search(tests, short_msg)` the node is not reported.
            You can use e.g. `tests=aux-chain|cop-upos` to apply only those two tests.
            Default = None (or empty string or '.*') which all tests.
        skip: a regex of tests to exclude.
            If `re.search(skip, short_msg)` the node is not reported.
            You can use e.g. `skip=no-(VerbForm|NumType|PronType)`.
            This has higher priority than the `tests` regex.
            Default = None (or empty string) which means no skipping.
        max_cop_lemmas: how many different lemmas are allowed to have deprel=cop.
            Default = 2, so all except for the two most frequent lemmas are reported as bugs.
        """
```

UD provides an official validation script. But Udapi (the Python framework to work with UD treebanks) now also includes a class that would look for violations of a series of hand-crafted rules

```
consistency (master*) » ./check_consistency.sh                                  francesco.mambrini
2024-11-28 22:16:17,443 [   INFO] execute - No reader specified, using read.Conllu
2024-11-28 22:16:17,443 [   INFO] execute - ---- ROUND ----
2024-11-28 22:16:17,443 [   INFO] execute - Executing block read.Conllu
2024-11-28 22:16:17,529 [   INFO] execute - Executing block ud.MarkBugs
2024-11-28 22:16:17,654 [WARNING] after_process_document - ud.MarkBugs Error Overview:
         degree-upos           1
            cop-upos           2
          case-child           3
          mark-child           4
           mark-upos           4
    conj-rightheaded           5
          multi-subj           9
          punct-upos          31
         punct-child          79
         no-PronType         100
           multi-obj         102
         no-VerbForm        3061
               TOTAL        3401

2024-11-28 22:16:17,654 [   INFO] execute - Executing block write.Conllu
```

Errator: an implementation of the annotation variation principle

1. Given one or several corpora of annotated sentences, find all *maximal repeats*, that is to say a substring that occurs at least in two different sentences and cannot be extended to the left or to right to a longer substring. The *maximal repeat problem* can be solved efficiently[1] using Generalized Suffix Tree (Gusfield, 1997).

2. For all repeats, extract their corresponding annotation following alignment links;

3. if not all the annotations corresponding to a single repeat are identical, flag them as a potential annotation error.

Wisniewski 2018

6	6	μή νυν	True	tlg0011.tlg002.perseus-grc2.tb.xml@2900439 μή νυν ἓν ἦθος μοῦνον ἐν σαυτῷ φόρει , ὡς φῂς σύ , κοὐδὲν ἄλλο , τοῦτ' ὀρθῶς ἔχειν . tlg0011.tlg003.perseus-grc1.tb.xml@2383722 μή νυν ἄτιμα θεούς , θεοῖς σεσωσμένος .
6	9	ἦ μάλα δή	True	tlg0012.tlg001.perseus-grc1.tb.xml@2281395 ἤθεἰ ἦ μάλα δή σε βιάζεται ὠκὺς Ἀχιλλεὺς ἄστυ πέρι Πριάμοιο ποσὶν ταχέεσσι διώκων · tlg0012.tlg001.perseus-grc1.tb.xml@2277683 ἆ δειλ' ἦ μάλα δή σε κιχάνεται αἰπὺς ὄλεθρος . 2185741 " Τηλέμαχ' , ἦ μάλα δή σε διδάσκουσιν θεοὶ αὐτοὶ ὑψαγόρην τ' ἔμεναι καὶ θαρσαλέως ἀγορεύειν ·
5	17	ἐπὶ νηυσὶν Ἀχαιῶν	True	tlg0012.tlg001.perseus-grc1.tb.xml@2278560 ἀτὰρ Τρώων κορέεις κύνας ἠδ' οἰωνοὺς δημῷ καὶ σάρκεσσι πεσὼν ἐπὶ νηυσὶν Ἀχαιῶν . 2186410 ἄλλῳ δ' αὐτὸν φωτὶ κατακρύπτων ἤισκε , δέκτῃ , ὃς οὐδὲν τοῖος ἔην ἐπὶ νηυσὶν Ἀχαιῶν .

Ex 6: the emphatic particle νυν with negative imperative is once joined with the negative particle, once with imperative. **Ex 6**: the sequence of particles ἦ μάλα δή is attested 3 times with 3 different annotations! **Ex 5**: in the Homeric formula ἐπὶ νηυσὶν Ἀχαιῶν ("to the ships of the Achaeans"), "Achaeans" is once adjective, once noun.

A practical exercise

- We take a **Greek** (or Latin) **text**
- We use a **pipeline available online** (UDPipe)
- We download the **treebank file**
- We load it into a **treebank editor**
- We **inspect** it (search it, edit)
- (We run the **UD official validation** to inspect problems)